

REQUEST FOR QUOTATION

Stepper Motor Bracket, SB-1042 Rev A

CNC machined 6061-T6 aluminum, clear anodized. Three quantity breaks.

RFQ number	RFQ-2026-0417	Issue date	25 Sep 2026
Buyer	Northwind Robotics, 1200 Harbor Ave, Unit 4, Portland, OR	Quote due	09 Oct 2026, 17:00 PT
Contact	A. Rivera, Mechanical Engineer procurement@northwind.example	Validity	Quote valid 60 days
Incoterms	FCA supplier dock (or DAP Portland, state which)	Payment	Net 30

Line items

Line	Part no.	Description	Material / finish	Qty	Target delivery
1	SB-1042 A	Stepper motor bracket, NEMA 17	6061-T6 / Type II clear anodize	25	6 Nov 2026
2	SB-1042 A	Stepper motor bracket, NEMA 17	6061-T6 / Type II clear anodize	100	20 Nov 2026
3	SB-1042 A	Stepper motor bracket, NEMA 17	6061-T6 / Type II clear anodize	500	Blanket, 4 releases

Scope of work

[AI] Supply finished, inspected parts per the enclosed specification sheet and 3D model. Scope includes raw material, CNC machining, thread cutting, deburring, anodizing, first article inspection, material certification and packaging. Quote each quantity break separately, with any one-time tooling, fixturing or programming (NRE) shown on its own line.

Package contents

Doc	Title	Format
A	RFQ cover (this page)	PDF
B	Part specification sheet and drawing views	PDF, p.2
C	Quality, finish and packaging requirements	PDF, p.3
D	Supplier quote response form	PDF, p.4
E	3D model, SB-1042_RevA.step	STEP AP242 (attached separately)
F	Data provenance and open items (internal)	PDF, p.5

How to respond

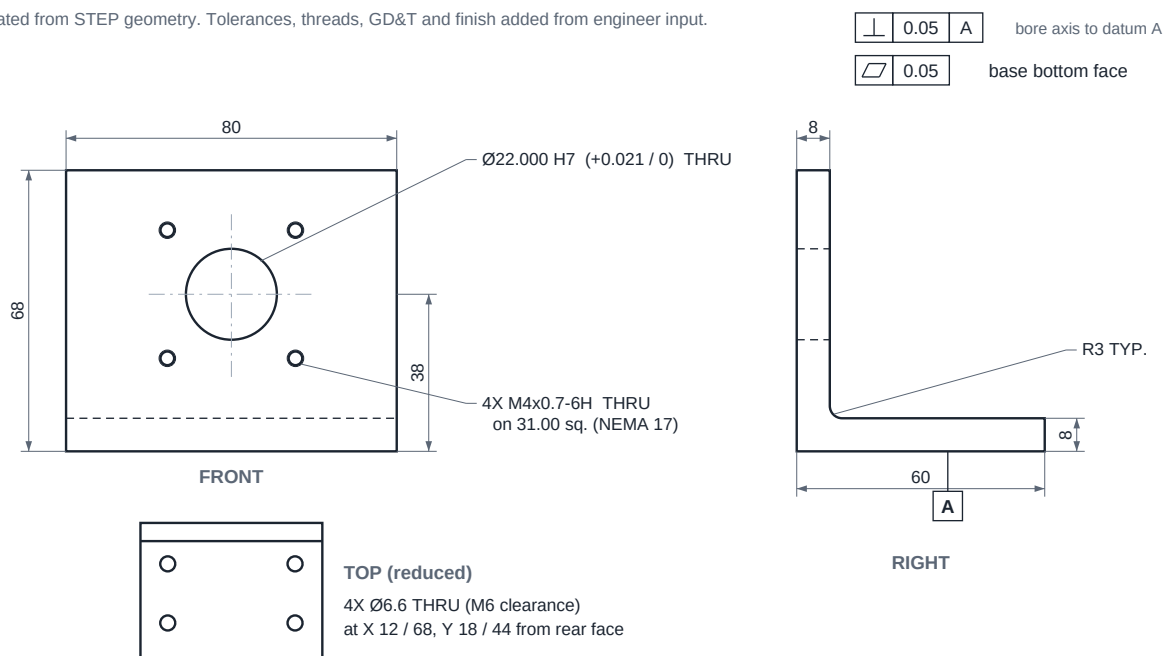
[AI] Complete the quote response form (Doc D) and return it with any exceptions to procurement@northwind.example by the due date. Questions go to the same address by 03 Oct 2026; answers are shared with all invited suppliers. If a requirement drives cost (for example the H7 bore or the perpendicularity callout), say so and offer an alternative. We review price, lead time, quality record and responsiveness, not price alone.

Field sources in this sample: [CAD] extracted from STEP geometry · [ENG] entered by the engineer · [AI] drafted by AI, reviewed by engineer. Tags would be hidden in the version sent to suppliers.

DOC B

Part Specification Sheet

Views generated from STEP geometry. Tolerances, threads, GD&T and finish added from engineer input.



STEPPER MOTOR BRACKET Northwind Robotics (sample company)	PART NO. SB-1042	REV A	MATERIAL AL 6061-T6	FINISH Anodize II, Clear
	UNITS mm	SCALE NTS	GEN. TOL. ISO 2768-mK	PROJECTION Third angle

Attribute	Value	Source
Overall envelope	80.0 x 60.0 x 68.0 mm	[CAD]
Volume / est. mass	72.8 cm³ / 197 g (at 2.70 g/cm³)	[CAD]
Stock size (suggested)	85 x 75 x 70 mm block, or 80 x 80 x 8 mm angle extrusion cut to 60 mm	[AI]
Features detected	1x Ø22 bore, 4x Ø3.3 holes (tap drill), 4x Ø6.6 holes, 1x R3 internal fillet	[CAD]
Material	Aluminum 6061-T6 per ASTM B209 (plate) or B221 (bar)	[ENG]
Threads	4x M4x0.7-6H through (not modeled in STEP; holes are tap-drill size)	[ENG]
Critical dimension	Ø22.000 H7 (+0.021 / 0) bore, motor pilot fit	[ENG]
GD&T	Perpendicularity 0.05 of bore axis to datum A; flatness 0.05 on datum A	[ENG]
General tolerance	ISO 2768-mK (±0.2 on 30 to 120 mm)	[ENG]
Surface finish	Ra 3.2 µm general, Ra 1.6 µm in bore	[ENG]
Finish	Anodize MIL-A-8625 Type II, Class 1 (clear), 8 to 12 µm. Mask nothing.	[ENG]
Likely process	3-axis CNC mill, 2 setups; bore finished by boring or reaming	[AI]

DOC C

Quality, Finish and Packaging Requirements

Critical-to-function characteristics

#	Characteristic	Requirement	Inspection method	Sampling
1	Bore diameter	Ø22.000 to 22.021	Bore gauge or CMM	100%
2	Bore perpendicularity	0.05 to datum A	CMM	FAI + 1 per 25
3	Base flatness	0.05	Surface plate + indicator	FAI + 1 per 25
4	M4 thread pitch	6H go / no-go	Thread plug gauge	100%
5	Hole pattern	31.00 sq. ±0.1	CMM or optical	FAI + 1 per 50

Quality documentation

- **First article inspection** to AS9102 format (or equivalent ballooned report) on the first part of each lot. Ship the FAI report before or with the lot.
- **Material certificate** (mill cert) showing alloy, temper and heat/lot number, traceable to the parts.
- **Certificate of conformance** signed by the supplier's quality representative.
- **Anodize certificate** from the finisher stating specification, type, class and coating thickness.
- Non-conforming parts are not to be shipped without written deviation approval from the buyer.

Workmanship and finish

- Break all sharp edges 0.2 to 0.5 mm unless noted. No burrs in holes or threads.
- Anodize after all machining. Threads may be chased after anodize; bore H7 applies after anodize.
- Uniform clear appearance. No stains, rack marks visible on cosmetic faces (front and top), or tooling marks deeper than Ra 3.2.
- Part mark SB-1042 A and lot code on the rear face with laser or vibro-etch, 2 mm characters. No ink.

Packaging and delivery

- Individually bagged or separated by foam so anodized faces do not touch. Bulk loose packing is rejected.
- Label each box with part number, revision, quantity, lot code and PO number.
- Include packing list, C of C, material cert and anodize cert in the first box.

Commercial terms

[AI] Drawings, models and this RFQ are confidential and remain the property of Northwind Robotics. They may be shared with sub-tier processors (such as the anodizer) only as needed to quote or produce the part. Changes to material, process or sub-tier suppliers after award require written notice.

DOC D

Supplier Quote Response Form

Return by 09 Oct 2026. All prices in USD.

Supplier name		Quote ref.	
Contact / email		Date	
Certifications	☐ ISO 9001 ☐ AS9100 ☐ ITAR registered ☐ Other: _____		

Pricing

Line	Qty	Unit price	Extended	Lead time (days ARO)	Notes
1	25				
2	100				
3	500				
NRE	1				Programming, fixtures, gauges
		Anodize incl.?	☐ Yes ☐ No	Freight incl.?	☐ Yes ☐ No

Process and capacity

Material source	
Machining process / setups	
Anodize done by	☐ In house ☐ Sub-tier (name): _____
Inspection equipment	☐ CMM ☐ Bore gauge ☐ Thread gauges ☐ Other
Monthly capacity for this part	

Exceptions, DFM suggestions and cost drivers

Authorized signature	Name and title	Date

DOC F · INTERNAL, NOT SENT TO SUPPLIERS

Data Provenance and Open Items

What the platform could pull from the STEP file, what the engineer had to add, and what the AI drafted. A plain STEP (AP203/AP214) carries geometry only; tolerances, material, finish and threads must come from the engineer, a drawing PDF, or AP242 PMI if the CAD tool exported it.

From STEP geometry	From engineer input	Drafted by AI
• Overall envelope 80 x 60 x 68 • Volume and mass • Hole count, diameters, positions • Bore Ø22 location • Wall thickness 8, fillet R3 • Orthographic views • 31 mm square hole pattern (recognized as NEMA 17)	• Material and temper • Finish spec and class • Thread callouts (M4-6H) • H7 fit on bore • GD&T and datum scheme • Surface roughness • Quantities and dates • Quality docs required	• Scope of work text • Response instructions • Suggested stock size • Likely process routing • Inspection method per CTF • Packaging and workmanship notes • Commercial terms paragraph

Flags raised during generation

Flag	What the platform noticed	Resolution
Resolved	Four Ø3.3 holes are tap-drill size for M4 but no thread data is in the STEP.	Engineer confirmed M4x0.7-6H through.
Resolved	Ø22.00 bore with 31 mm square pattern matches NEMA 17 motor pilot.	Engineer set H7 fit and perpendicularity.
Resolved	Anodize adds 4 to 6 µm per surface; H7 bore is only 21 µm wide.	Note added: H7 applies after anodize.
Open	Inside corner R3 with 8 mm walls; machining from solid block wastes about 85% of the stock.	Ask suppliers to quote angle extrusion as an option (form has a DFM field).
Open	No part marking location modeled.	Rear face proposed; engineer to confirm.

Time to produce

Estimated for this sample: upload and geometry extraction under 1 minute; engineer fills about a dozen prompted fields in about 10 minutes; AI drafts text and layout; engineer reviews and exports. A typical manual RFQ package for a part like this takes an engineer 1.5 to 3 hours (estimate, not measured).